Economists and Development: Rediscovering Old Truths

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A spate of economic development literature appeared in the late 'fifties and early 'sixties as the advanced nations' governments and economists turned their attention to the so-called developing nations. A recent survey of subsequent contributions to the theory and practice of development reveals less that is new and important. Meanwhile poverty persists in the less developed countries (LDCs). And economists are realizing that *several* social sciences are involved. What else has been discovered or relearned during the "decade of development" now drawing to its close?[1]

Theoretical Contributions Limited

Economic development remains one of the most unstructured branches of academic economics. It does not offer much of interest to those who treat economics as a form of applied mathematics. The subject has attracted not quantitative economists, but those concerned with institutional and social problems. During the late fifties many of the greater names in economics turned their attention briefly to economic development questions. Some of them have since returned to other interests and few have sought to implement their ideas or take development assignments abroad.[2]

The very considerable cultural variations that exist among continents and even neighboring countries have often inhibited the sort of abstract and deductive theorizing so characteristic of other areas in economics. The extensive literature of particular description also distracts attention from several typical and important circumstances (*e.g.,* unchecked human fertility). Economic and social theorizing if applied to more of these circumstances in common should advance the understanding of policy makers and others.

Unfortunately the two main theoretical contributions of the inter-war period have very limited relevance to LDCs. Principles of monopolistic (or "'imperfect") competi-

[1] This essay is intended as a review of the economic development "state of art" as regards both practice and theory. Because of space constraints, and because this article is not exclusively a *literature* survey, it cannot mention all important published contributions. The author's views are based in part on experience gained on contract development projects in and for LDCs and from work with AID on and off over the years.

[2] Many of the really seminal ideas concerning the development of LDCs were published before 1965. Later publications have tended to be extensions of these same ideas. The last five years or so evidence a slower advance in understanding.

tion filled a large gap in value theory, but product differentiation and advertising relate far more to advanced countries.[3] Keynesian theories of liquidity preference and employment-output determination in the short run apply only obliquely to the partial subsistence economies of most backward countries.

Since 1950 a variety of so-called "growth models" have been presented by Meade and others [49. Meade, 1961; 58. Qayum, 1964; 59. Ranis and Fei, 1961]. An important subset of these are extensions of the intellectually elegant but seriously limited Harrod-Domar interactions of aggregate savings and capital productivity rates [32. Harrod, 1939; 18. Domar, 1947]. Among the more useful are those that include the output consequences of more available labor plus the interacting effects of varying capital to labor ratios on worker productivity and unemployment rates. More recent growth models also allow in various ways for improvements in the technological state of art. Modified applications of Cobb-Douglas models, especially by Solow, exemplify some of these attributes [69. Solow, 1961; 70. Solow, 1956; 4. Amano, 1964].

When such growth models are combined with demographic models something can be learned about the incidence of changing birth and death rates upon future national outputs, incomes per head, rates of return on capital, etc. [22. Enke, 1969].

Another approach is offered by the "two gap" model of H. B. Chenery [13, 1966], in which the domestic savings gap and the available external capital gap are related, the latter allegedly being the more serious constraint during early stages of development. Models of this kind recognize the additional difficulties that exist in practice when equilibrium "prices" such as foreign exchange rates are not permitted. They seem to imply, as do other "capital" models, the labor either is applied in fixed proportions with capital or has a zero marginal product.

Of current interest are multi-sector models that differentiate between public and private sectors, urban-industrial and rural-agricultural sectors, and possibly an export sector. Here a prime requirement is an acceptable explanation of rural-urban migration. These models are truly useful when they show the necessary relation between increased productivity and output in both agricultural and industrial sectors [74. M. Todaro, 1968; 75 M. Todaro, 1969].

Since World War II there has been at least one publicized "historical" theory of economic development by W. W. Rostow [61. 1956; 62. 1959]. The "stages of growth as defined, and more especially the announced requirements for an underdeveloped country to "take-off" into sustained growth, attracted considerable attention for a while. Subsequent critiques have reduced acceptance of this viewpoint [20. Drummond, 1961; 8. Cairncross, 1961; 42. Kuznets, 1963].

Many perceptive economists have recognized the special role of profitable innovation in economic development. Schumpeter's classic really applies to backward as well as advanced countries. Unfortunately there are not enough descriptions of agricultural and industrial innovations in LDCs [36. Johnson, 1960; 37. Jones, 1957; 48. Marriot, 1952].

Some of the theories of economists that have helped and hindered a better understanding of the development process are outlined below. A simple but useful schema for viewing the development processes continues to be that of Lewis, for example, with labor initially in unlimited supply, but workers subsequently being combined each

[3] This does not imply that limited monopolies and imperfect knowledge do not abound in LDCs. The contrary is true because of few domestic producers, obstacles to importation, and high internal transportation costs that further subdivide the national market. It is the element of deliberate brand differentiation that is missing.

with more capital [45. Lewis, 1954; 46. Lewis, 1958; 23. Enke, 1962]. Several of these ideas together comprise a useful but loosely related body of doctrine.

Nevertheless, no integrated, explicit, and unique theory of development exists as of today. The nature of the subject may preclude any such abstract and complete analysis. It is significant that no single work on development has yet appeared that is markedly superior in its insights to Adam Smith's *Wealth of Nations* (1776).[4]

Agricultural Revolution Necessary

A common feature of agriculture in LDCs is its relative inefficiency as measured in the high fraction of population engaged in producing food to provide even a meager diet for all. Yet most LDC governments associate industrialization with development and hence favor an expansion of industrial output that exceeds the ability of a neglected agriculture to support it. However, although often disregarded, something can and has been said by economists regarding conditions of equilibrium between these two sectors [34. Johnson, 1955; 66. Singer, 1964; 27. Flanders, 1969].

Accepting Engel's "law," there must be a *general* increase in productivity of labor and capital so that a rising output and income per head will result in a larger demand for outputs of industry relative to those of agriculture. But industrial output will not in fact increase relative to agricultural output unless there is a shift of labor and capital to industry from agriculture. This may be associated with an increase in productivity in agriculture *relative* to that in industry that "releases" labor from farms to factories. Often this will also be reflected in a relative decline in agricultural product prices. This permits industrial workers to buy all the food they wish to buy with less of their wages. It provides an extra incentive to immigrate to industrial centers.[5]

In many LDCs this relative increase in agricultural productivity has not occurred, partly from neglect by government but also because of fragmented holdings, lack of capital, and inappropriate techniques. Lacking has been anything like the subsidized experimentations and county-agent-type assistance to farmers so long characteristic of the United States. Temperate zone agriculture has been incorporating one technological advance after another for the past two hundred years, and is continuing to do so, with United States agricultural productivity increasing at 4 percent or so annually. Most LDCs are located in or near the tropics; their agriculture often has to cope with quite different soil and rainfall conditions, while in many areas the need for village farmers to farm alike discourages independent innovation.[6]

Rapidly increasing populations are making some LDC governments worry more about agriculture because of extra food needs. But additional food supplies are only a partial solution to malnutrition. GNPs must also increase and comprise more industrial goods because otherwise extra food output will not be exchangeable in cities. In a closed and growing economy there must be extra food produced by agricultural workers *and* demanded by industrial workers. Intersector "balance" is always a requirement.[7]

[4] See Spengler [72, 1959]. Those who treat industrialization almost as synonomous with development should remember that the *Wealth of Nations* was being written before the main onslaught of the industrial revolutions but after many important agricultural innovations. The closest modern analogue to Smith's classic is probably W. Arthur Lewis' *Theory of Economic Growth.*

[5] See Ranis and Fei [59, 1961] for an elegant description of the development process as it relates to transfers from agriculture to industry.

[6] See Jones [38, 1961], Southworth and Johnston [71, 1967], and Schultz [65, 1964] for a description of certain ills and woes of traditional agriculture in the tropics.

[7] See the recent Presidential report on food and population [57, 1967] in which the importance of

The developed nations together are providing a comparatively small fraction on balance of the LDCs food consumption. In exchange they are accepting petroleum, minerals, certain agricultural products, and restricted payments. To this limited extent the agriculture of the Western world, especially through wheat and rice shipments, is providing a significant food base needed for industrialization by LDCs. Perhaps, if the newly settled temperate zones have a comparative advantage in modern food production, this can and should endure indefinitely. However it would seem that true economic development requires that the LDCs eventually have an LDC agricultural revolution, analogous to that which still accompanies the continuing industrial revolution in North Western Europe and North America. Agricultural innovation remains a prerequisite of industrialization.[8]

Capital is Insufficient

A superficial view, but very prevalent *circa* 1960, was that infusions of "capital," if large enough, are alone sufficient to induce "development." This doctrine became instantly popular not only with bureaucracies that administer loans and grants but also with government officials of recipient nations. It was conveniently forgotten that capital is always invested in a particular socio-economic context that largely determines its productivity in use.

This Big Push "capital is everything" doctrine assumed all past increases in a country's GNP to be attributable to increases in its domestic capital stock and *not* to temporarily associated increases in labor employment or improvements in technology for instance. Estimations of ICORs, or incremental capital-to-output ratios, were made for various countries during the past decade or so. Then, if historically each extra $1 of output was associated with $3 of extra domestic investment, it was supposed that $3X of *further* additional "'capital" from abroad would somehow generate a *further* additional $X of national output [60. Rosenstein-Rodan, 1961].

Inherent in this technocratic arithmetic were neglect of capital absorptive capacity, such as the availability of related skilled labor and effective demand for the output, plus some confusion as to the nature of capital itself. Even more glaring was neglect of substitution effects between the "capital" loaned and private and public consumption in the receiving country. A large transfer of international purchasing power to an LDC government, however closely tied to specific physical investments, can still result in that government taxing less and selling exchange more than it otherwise could or would. Funds and resources are more fungible than most administrators and some economists care to admit.[9]

A related and weird inversion of fact concerned how to generate sufficient purchasing power to buy domestically the output of new industries in an LDC. It was argued that, as the workers of a new industry H would not buy all *its* output (sic), new industries H through Z would have to be created with a big push investment so that their collective outputs could more nearly be purchased at cost by their workers together. (Realistically, a demand for

effective market demand for food was stressed thanks to the insights of Eric Thorbeke.

[8] A related subject is the changing industrial structure of developing economies. One of the most thorough treatments is by Chenery and Taylor [14, 1968]. Some of the pioneering work has been by Kuznets [41, 1957; 43, 1966].

[9] The simplest and most erroneous view is that $X of loans or grants represents a net increment in "capital," usually visualized as industrial plants, which using free labor almost automtically yield extra output without regard to effective demand for it.

industry H arises when productivity improvements in industries A through G enable their workers to spend a large enough fraction of their wages to buy the H output, which is just the opposite of the Big Push view.)[10]

Today it is recognized that capital accumulated from whatever source is not enough. It is extra capital with extra labor and new processes that ordinarily mean extra output. The notion that in LDCs the marginal productivities of capital and labor are respectively very high and low relative to advanced countries is now less widely held. Nevertheless, this remains an important research question of some urgency, for if these productivities were better known they should affect the priorities of lending and granting agencies.

In very densely populated areas the marginal product of farm labor may be low because of scarce arable land. But most LDCs also include large regions of sparsely populated but not infertile land. Then there may be several ways in which government could increase the output of a "last" hour worked. If the worker's output is low because of ill health (*e.g.*, malaria) or inappropriate practice (*e.g.*, not using fertilizers) there are obvious remedies available. Marginal labor productivity is not a *datum.* It can be raised through human capital investments and technological experimentation. A new cultural pattern, with a greater play of financial incentives, is also often necessary.

Reduced Fertility Essential

For LDCs the "Decade of Development" has typically resulted in GNP increases of 5 percent or so and population increases of 2.5 percent or so annually. Probably half of GNP increases have been "swallowed" by population increases. This record of two steps forward and one backward has increasingly focused attention on birth control programs as one of several necessary economic development measures.

Only during the past decade have development economists become aware of the favorable incidence on per capita incomes of decreases in human fertility rates. Demographers have long known that dependent children become a smaller fraction of the population as age-specific fertility declines, but it was the pioneer work in 1958 of A. J. Coale and E. M. Hoover that related this in turn to increased savings, investment, and GNP per head for India [15]. Later this approach was carried further by Hoover and M. Perlman [33. 1966] and others.

Starting in 1960, Enke has argued that births are excessive from a purely economic (non-parental) viewpoint when infants at birth have a negative discounted value [24]. He has also argued that where births are excessive it is many times more economical to raise per capita incomes through slowing population growth rather than further accelerating output through traditional investments in factories, irrigation, infrastructure, etc. [25. 1966]. Recently, using combined demographic-economic computer simulation models, there have been several dynamic analyses that largely confirm conclusions based on static analyses presented before [56. United States Population Council, 1965; 78. Zind and Enke, 1969].

These analyses have also shown that it is not absolute population *size* so much as population growth *rates* that lessen improvements in GNP per head. If an LDC's population is doubling in 25 years, the natural resource endowment being fixed by definition, income per capita must fall unless there is an offsetting combination of accumulated investment and technological advance. Ordinarily both of these elements

[10] The initial misstatement of the real problem seems to have come from Nurkse [53, 1952], but it was Rosenstein-Rodan who publicized the idea that the output of a new industry should be bought by its productive factors' incomes rather than from those of old industries.

need time to come adequately to the rescue. Most LDCs could have populations several times larger at *some* future date. Many of them contain large and comparatively uninhabited areas that have remained sparsely populated for some reason. However, whatever the national population objective, it had better be achieved later rather than sooner if per capita incomes are to rise rather than fall.

The obvious consequences of natural population increase, quite apart from these abstract economic analyses, has meanwhile prompted several governments, including India, Korea, Pakistan, Taiwan, Tunisia, and Turkey, to undertake large scale contraception programs. In the last few years many other governments outside Latin America have begun modest programs. United States AID since 1965 has been increasingly active in providing assistance on request. The World Bank's presidents, especially McNamara, have stressed the need to provide contraceptive assistance on a voluntary basis if rapidly rising populations are not to significantly offset the modest GNP increases of most LDCs.

Whether voluntary contraception programs of the kind now widely favored can be sufficient and effective has been seriously questioned by K. Davis [17. 1967]. It has also been indicated by Kuznets [44. 1967] that there is always some increase in saving propensities or technological innovations that gives the same GNP/Population increase as does a given fertility reduction. Operationally, and more important than arithmetic equivalence however, is that the sort of adult couples that do not practice birth control are not *therefore* likely to save and/or innovate more vigorously.[11]

[11] The opposite might be argued even more forcibly, namely, that the sorts of parents who have the foresight and discipline to practice contraception effectively may be more rather than less likely to save and innovate for their future material well-being.

Investments in Education and Health

National resources can often be invested advantageously in improving people as well as in accumulating physical capital—in fact one is not much use without the other. Quality of people—including their education and health—can be more important than quantity of population. A prevalence of vigorous, informed, and socially responsible people is proof of successful economic development.

That education and health can be a very profitable investment was recognized long ago by a few economists [52. Nicholson, 1891]. Parents often invest *in* their children rather than for them. But T. W. Schultz was one of the first economists in the post war era to stress the high rates of return obtainable from investments in human capital [63. 1960; 64. 1961]. He and others have shown that in certain situations the aggregate true costs of education are a profitable investment for the individual, his dependent family, and government collectively. Others have considered possible trade-offs, such as between a modest education for many (*e.g.*, general literacy) as against more intensive training of a few (*e.g.*, able students are subsidized through college), besides considering the more general trade-off between education and, say, industrial investment [5. Anderson and Bowman, 1965; 7. Becker, 1964; 76. Weisbrod, 1961].

Obviously, the kind of education obtained will influence the rate of return, and the kinds of education that most increase future earnings streams will vary among people and countries. In many LDCs there is a tendency to avoid vocational training that includes manual work and to stress unduly a traditional education for clerical positions that are too few.[12] Education as a

[12] A serious criticism of colonialism in this century is the adoption of curricula in primary and secondary schools in backward colonies suitable

means to achieve development is often too serious a matter to be left exclusively to professional educators.

A serious counter-loss to many LDCs is the brain drain that results from university training abroad.

Returns on investments in health, whether to the individual or society, are much harder to estimate. How vigorously a man works, be it manual or clerical, is ordinarily related to his state of health. Mañana may not be a state of mind but a state of ill health. On this score public health expenditures to prevent and cure the crippling diseases can usually be justified on economic grounds alone. Examples are programs against malaria, bilharzia, hookworm, sleeping sickness and adult gastro-enteritus [1. Abel-Smith, 1967; 77. Winslow, 1951; 51. Mushkin, 1958].

More controversy surrounds the economic, as distinct from human justification, of expenditures to prevent such killing diseases as cholera, plague, and smallpox, despite their low costs per individual benefited. Clearly, the premature death of a recent college graduate represents a serious economic loss as well, but is this true of infant mortality? Indeed, some have even questioned expenditures to reduce infant mortality, especially at a time when resources are also being diverted to contraception in some countries. There should be no conflict however. The more widespread are contraception (and possibly abortion), the more wanted by their parents are those children that are born, and the better will be their education, health, and general care.

Expenditures against premature death serve the objectives of true development. Families are more willing to invest and innovate when their survival to enjoy the results is more certain. Risks of early death from disease must inhibit a family's planning to advance itself.

Nevertheless, as public health programs reduce mortality rates, the burdens of rapid population growth do become more serious. But the remedy is not to limit public health. It is to supplement these health programs with voluntary birth control campaigns. Birth control and child care are consistent. The social purpose of medicine, whether to prolong life or prevent births, is to give men and women more control over their lives [29. GE-TEMPO, 1968].

Human capital investments are a force making for cultural change. Many economic attitudes alter when a long and full life becomes more assured. But more important, true education not only informs but trains a person to analyze, decide for himself, and to reject much that is superstition, customary, and unfounded.

Government Planning of Economic Development

Most LDCs have economic development "plans" that seek to establish priorities among expenditures on industries of various kinds, agriculture, different infrastructure possibilities, and education and health. Or, if an LDC government does not have such "plans," some outside agency such as United States AID may have less official ones. A basic but largely unanswered question is "How should priorities be established?"

There is a rather extensive literature on the methodology of establishing priorities that is ordinarily more conceptual than useful [21. Eckstein, 1957; 39. Kahn, 1951; 70. Solow, 1956; 47. Lewis, 1966; 40. King, 1966].

One logical difficulty is to decide what the "output" is that is to be increased; GNP, foreign exchange, or jobs, for instance, and in each case what social class or geographic area should benefit most? Another difficulty

almost exclusively for passing entrance exams into universities in London, Paris, etc.

is to decide what is the scarce input from which this output must be maximized: for example, is it government expenditures, net domestic savings, or a skilled labor category such as management? Among academic economists at least there is fair agreement that governments should not imitate private capitalists and make capital funds investments so as to maximize a financial rate of return. The crushing argument is always that there are "externalities," benefits that cannot or should not be captured as money receipts, and it is their existence that justifies not-for-profit investments in the public sector.

This viewpoint is obviously attractive to officials of international and national development agencies. Loans or grants for infrastructure or social overhead capital can be made with fewer worries if no significant receipts are expected from the investments and there can hence be no profit test. A loan to construct a shoe factory must prove itself profitable or unprofitable because shoes are sold. But a loan for a highway usually need not because the highway has "externalities" and tolls are unpopular. However, with equal imagination, a health argument could be also made for wearing shoes (*e.g.*, against hookworm), and highway user charges are not unknown.[13]

Even were there an accepted theory for establishing investment priorities, it would be hard to apply in practice. Cash and benefit flows have to be estimated for at least each project's duration. There is the usual question of what discount rate to use. Maintenance is uncertain. So is the availability of labor and the extent of the market. Planning usually has to be without facts [73. Stolper, 1966].

In determining *programs* rather than *projects,* any reduction in difficulty is more apparent than real.[14] The truly relevant facts or estimates remain elusive, and basic uncertainties regarding objectives continue. Without goals that are precise and detailed enough to have operational significance, there can be no real criteria. The truth is that officials of development agencies can do little more than lend and grant on judgment, experience, or hunch, and each will have a somewhat personalized view of what is necessary for other people's welfare. In a way this is what was always done by more enlightened colonial administrators. There may be no other way.

Several LDCs that have become politically independent since the war are nominally "socialistic" to some degree. This usually means that certain industries are owned and operated by the government that in "capitalistic" countries would comprise private corporations. Also this often means a Five Year Plan, with higher outputs as goals, and some related projections of government budgets, private savings, and firm and household expenditures. These planning exercises, which may include an input-output table, can be useful if projections and coefficients are realistic and the plan is not taken too seriously by officials, investors, or even voters.

In LDC economies, despite a plethora of regulations, the positive economic behavior of persons is still largely decided by themselves. An LDC government can no more enforce a plan to expand outputs than it can enforce a plan to reduce births. Behavior in each case must be a voluntary response to incentives.

[13] The present concept of "externalities" is very similar in some cases to the old Pigovian disparities between social and private product. Useful statements on "externalities" and economic development planning are Scitovsky [68, 1954], Fleming [28, 1955], and Fellner [26, 1956].

[14] Among development practitioners there has been a lively controversy on the pros and cons of program versus project assistance. But the decision does not relate so much to criteria and knowledge as to the warrantable degree of freedom over spending that should be given to LDC government officials. Useful references are Singer [67, 1965] and Carlin [10, 1967].

Mobilizing Resources Through Stabilizing Prices

LDC governments are not unique in wishing to spend more than they can tax or borrow—but they do have additional reasons for resorting to deficit finance. Collection of "productive" taxes (*e.g.*, on incomes) is administratively difficult, significant land taxes are often politically impossible, while easily collectible direct taxes (*e.g.*, import duties) have limited yields. Many commonplace ways by which advanced economies channel savings into investments—*e.g.*, life insurance companies—are hardly practicable in countries where prices are increasing 25 percent or so a year.[15]

The extent to which inflation is necessarily associated with development has at times been a subject of debate. Inflation of over 10 percent a year exists in many LDCs but such an association does not seem necessary either theoretically or historically. More excessive inflation is perceived and discounted, so that savers will not buy debts, while savings by those who wish to avoid risks yet cannot themselves be entrepreneurs are discouraged [35. Johnson, 1966; 31. Harberger, 1964; 50. Mundell, 1965].

Increasingly recognized also is that excessive inflation leads to price ceilings and government allocations of necessities—such as rice and housing in cities—frustrations of the price system which eventually result in more serious shortages and costs.

Has Government Interference Been Excessive?

It is not clear whether certain LDCs have developed economically despite or because of economic participation and regulation by their governments. A crucial issue is still the degree of government involvement in economic affairs that most promotes development. Any opinion must take into account the capacity of government to plan and manage wisely given the average quality of people it can employ.

Acknowledging wide variations, some LDC governments now try to regulate the economic behavior of individuals to a degree that has been unknown in the Western world since the days of mercantilism. Foreign exchange prices and transactions are controlled. Imports and exports are taxed and limited by quota. Internal movements of grains and other commodities are sometimes regulated. The establishment of many new businesses is by license. Investments to expand output capacity may need approval. Controlled also are rentals of apartments and houses, the prices of certain basic "commodities," and the wages of various labor skills. The distribution of basic "necessaries" may be under ration. It would be serious indeed if all these regulations were really enforced. But the mere existence of these regulations and penalties for violating them does inhibit private enterprise while encouraging corruption.[16]

Although such statements are unfashionable, it must be recognized that all bureaucracies, unless prevented, will impose regulations to increase their staffs and to conceal policy mistakes. For many officials the existence of regulations means an income from bribes. Also, in many LDCs corruption is a heavy "tax" on enterprise, and when combined with excessive regulation this results in a flight of foreign-owned capital whenever possible.

[15] An additional inhibitor, not always understood by Westerners, is that loyalties in LDCs are to the family, caste, subtribe, etc., rather than some fiduciary virtue. Thus joint stock companies are usually family affairs that cannot attract the savings of persons outside the group. In LDCs it is often necessary for substantial savers and entrepreneurial investors to be related by blood or marriage.

[16] Bauer [6, 1959] describes several frightening examples of the degree of government controls permissable and/or practised in India.

In every country there are a relatively few people, often immigrants of different culture, who are materialistic, experimental, frugal, industrious, and ambitious for their families. These few by innovating, saving, and employing more and more others, have in the past always been the agents of true economic development. But they are less likely to emerge if governments harrass them and an established class structure suppresses them.

Some LDCs are governed by military men, doctors, or lawyers. Others are governed by intellectuals with Fabian tendencies. Few of them are governed by men with successful experience in commerce, agriculture, or industry. The staffs of government agencies ordinarily come directly from school or college. Under these circumstances, LDC governments should probably involve themselves less rather than more in their economies, and allow more rein to ubiquitous private forces of development.[17]

Loans or Grants or Trade?

The past decade has seen a shift of official development assistance from hard loans to effective grants. It has been increasingly difficult to find profitable investments at rising interest rates—another reason for the popularity of not-for-profit infrastructure loans. But the difficulty of servicing the accumulating indebtedness is also causing a relative substitution of grants and very soft loans for hard loans.[18]

For some LDCs debt service may have to be rescheduled so drastically as to constitute default. The post war flow of official funds—which excludes private direct investment—apparently has often resulted in neither adequate domestic returns nor adequate foreign earnings for transferring debt service. If for an LDC this continues year after year, it eventually becomes impracticable to service external debts. Moreover, once it becomes apparent to certain LDC governments that the annual "capital" inflow does not significantly exceed debt service requirements, one incentive to be a good creditor largely disappears. Official hard loans for investments that do not clearly earn sufficient foreign exchange—*e.g.*, most infrastructure projects—may almost have to cease within another decade.

Grants may increasingly become a condition and source of rescheduled debt service. Realization that taxpayers' funds are repaying private hard loans may in turn increase the political acceptance of increased commodity imports into the Development Assistance Countries. The New York, London, and Paris financial communities should also support this trend.

If LDCs increased commodity exports to the DAC, in response to tariff concessions by the latter, the polemic literature may again be filled with theoretical "demonstrations" that the barter terms of trade always worsen secularly. Again there will no regard to increased efficiency in export mining and agriculture or to quality improvements in the industrial imports received in exchange. And again it will be implied that LDCs curiously have no comparative advantages [12. Chenery, 1961; 11. Chambers, 1966; 16. Cohen, 1968].

The gradual substitution of grants for loans may be accelerated by a growing realization that national credit worthiness and high birth rates are inversely related. At least one simulation model indicates that

[17] It is unfortunate that most financial assistance is government-to-government. Unreconstructed free enterprisers would like to see more public funds flowing into private entrepreneurial use through development banks. This would be an obvious use for PL 480 counterpart funds were it not for the objections of LDC officials who wish these funds to remain immobilized so as to lessen inflation and competition for resources.

[18] See OECD [54, 1967] and [55, 1968] for an impression of the volume and changing composition of official assistance and other capital flows to LDCs.

the sustainable annual improvement in GNP per capita, at least for several decades, is extremely sensitive to human fertility rates. If either the GNP per capita improvement aspiration is "too" high (so that additional external borrowing becomes necessary), or birth rates are "too" high, an LDC can never repay its external debts even without interest [30. GE-TEMPO, 1970].

Cultures and Institutions Must Change

There will never be real economic development in most LDCs without extensive cultural change and an erosion of "dualism."

It has taken many "Western" economists a long time to discover this truth. Thus it used to be thought that, because the Marshall Plan succeeded in Europe, large capital loans would bring economic development to poor and backward countries. It was forgotten that Europeans were the very originators of free enterprise capitalism and required no change of culture merely to rebuild industries destroyed by war.[19]

Many LDCs have cultures of historical importance and human greatness. But they do not promote innovation and other necessary efforts to produce more goods and services. The sacred cow in India, despite its value as a dung-fuel producer, is a commonplace example. But more important, caste systems reduce occupational changes, and class hierarchies prevent vertical mobility. Agriculture is too often a community activity so that an entire village must agree before a new cropping pattern can be adopted. Filial piety hinders migration. A cultural stress on fertility mitigates against contraception [2, 3. Adelman and Morris, 1965, 1968; 19. Domar, 1966].

The peoples and governments of LDCs resist the cultural changes that are associated with the economic development they want. Real economic progress is recognized as disruptive. People are forced off the land, occupations become obsolete, family authority is undermined, and new towns of strangers come into being that lack utilities and traditions of government.[20]

Therefore it is not surprising that government leaders of LDCs—who necessarily have short time horizons—often prefer loans and grants from agencies of the developed nations to promoting cultural change at home. But financial assistance despite "strings" often defers acceptance of change. In some instances financial assistance may work against technical assistance by lessening the need to incorporate new methods.

Rediscovering Old Truths

Many of the expectations and theories of the late 'fifties have since been reduced and amended respectively.

Both the practical and theoretical emphasis is now more on agriculture and less exclusively on industry, more on human and relatively less on physical capital investment, more on private enterprise and less on state industries, more on reducing birth rates and less on increasing national output, and more on cultural and technological change instead of making foreign loans and grants called "capital." The lack of real managers and technical specialists is seen as especially acute in the recently decolonized nations. The importance of efficient and honest government is becoming more appreciated.

Economists who practice development are also reverting to a much earlier role, becoming more concerned with alleviating

[19] It is also often overlooked that Western Europe *itself* saved all the capital it ever invested at home or abroad—there was no other region of the world from which to borrow.

[20] All of which undermines the "full-belly, anti-Communism" justification for appropriations often made to Congress by United States AID in past years.

poverty, the quantity and quality of population, and the incidence of local institutions and cultures. Many old truths are being rediscovered. Altogether the "Decade of Development" has been a sobering ten years [9. Cameron, 1967].

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